

Newton's Method

Section 4.6

A. The Newton's Method Formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Examples

1. **(Video)** Starting with $x_0 = 2$ find the third approximation x_3 to the root of the equation $x^3 + 5x - 12 = 0$

n	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0				
1				
2				

$x_3 =$

2. **(Video)** Starting with $x_0 = 0$ find the second and third approximation to the root of the equation $e^{-x} = 5 + 7x$

n	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0				
1				
2				

$$x_3 =$$

3. Starting with $x_0 = 1$ find the third approximation x_3 to the root of the equation $\tan^{-1}(x) = 1 - x$. Round (ONLY) your final answer correct to eight decimal places.

n	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0				
1				
2				

$$x_3 =$$

4. a. Find the equation $f(x)$ that results in a solution of $\sqrt[4]{9}$

b. Find the second, third, and fourth approximations of the root to this function if $x_0 = 2$

n	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0				
1				
2				
3				

$x_4 =$

